Predicting CYP Inhibition: Method Report

OpenADMET CYP Inhibition Blind Challenge (`GiantWizard`)

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1 Overview and Innovation Summary

This report covers entry `GiantWizard` in the OpenADMET CYP Inhibition Blind Challenge, covering direct inhibition regression across CYP1A2, CYP2C9, CYP2D6, and CYP3A4, and time-dependent inhibition (TDI) classification on CYP2D6 and CYP3A4. No proprietary data was used at any stage. All code and reproduction scripts live at github.com/GiantWizard/openadmet-cyp-challenge.

The regression track scores macro soft-threshold relative absolute error (MA-ST-RAE). A prediction that lands anywhere inside a compound's experimental credible interval costs zero error; outside the interval, the metric charges distance to the nearer edge. Those intervals are about ten times wider below $\mathrm{pIC}_{50} = 4$ (where potency sits below the lowest tested dose) than they are for strong hits.

The organizers describe the Innovation in ML Award as rewarding exploration over leaderboard tuning, judged on architectural novelty, creative use of available data, novel training or uncertainty quantification strategies, and simple yet effective engineering, independent of rank. This entry does not claim architectural novelty, since every model in it is a stock Chemprop, CheMeleon, TabPFN, or LightGBM. Instead, I tested how the assay design and the scoring function interact with standard models across the other three criteria:

Under **creative data usage**, the training table is a sparse matrix where compounds that failed the primary screen were never run in dose-response follow-up, while the 750-compound blind set is dense. Furthermore, as documented in the challenge announcement, the blind set was built by buying 10 Enamine catalog analogues each for the top 25 hits from CYP1A2, CYP2C9, and CYP3A4, while CYP2D6 was still in development on the Echo-MS platform. Relative to CYP2D6, most of the 750 test molecules are unexpanded non-inhibitors. Adding blank primary-screen rows as low-potency training examples fixed that gap and moved the entry from rank 55 to 41 across two uploads (`v26` and `v27`). Separately, buying top-hit analogues means the blind set sits at a median Tanimoto similarity of 0.578 to the training set, while a scaffold-disjoint holdout sits at 0.435. Standard cluster-disjoint cross-validation evaluates a problem that is much harder than the one being scored.

Under **uncertainty quantification**, training a model directly on the clipped interval loss and shifting its predictions afterward with an affine map turn out to be the same intervention reached two ways. Each helps on its own (0.7442 to 0.7244 before calibration), and combining them cancels out the improvement (reversing to 0.7060 for standard loss against 0.7198 for interval loss after

calibration).

Under **clever engineering**, dropping a CPU-era habit of compressing 605 features down to 200 principal components before TabPFN gained more than six weeks of blending and calibration changes, and was the only change that improved all five published leaderboard metrics at once.

My five-arm stack uploaded alone on 29 October (`v46`) scored **0.4472**. Averaging it with Scigantic's open-source submission R25 brought the score to 0.4335 the same day, and widening each column's interval spread without changing the mean or compound ordering brought the final submission (`v56`) to **0.4192**.

2 The Findings

2.1 A model trained only on follow-up hits has never seen a weak compound

The training CSV has blank pIC_{50} cells. The organizers confirmed in the challenge Discord on 8 October that a blank means the compound did not inhibit enough in the single-concentration screen (run in the +NADPH pre-incubation condition) to pass the log-fold change and Benjamini-Hochberg FDR gates for 12-point dose-response confirmation. The blind set, however, was assayed densely across all four enzymes.

For CYP2D6, this matters twice. First, CYP2D6 was measured by dextromethorphan depletion on a SCIEX Echo-MS system rather than the Vivid fluorescence kits used on the other three isoforms. Second, the 750 blind compounds were selected by hit expansion on 1A2, 2C9, and 3A4 before the 2D6 Echo-MS dataset was ready. Most of the test set is therefore inactive on CYP2D6.

In the training data, all 1,493 labelled CYP2D6 compounds respond at -0.59 or stronger ($\log_2$ fold change) in the primary screen. Of the 2,882 blank CYP2D6 rows that have screen measurements, only 5 reach -0.73. A model fitted to the numbered rows alone has never met an inactive compound. CYP3A4 and CYP2C9 behave differently: their blanks include compounds that responded strongly in the primary screen, so the same reading does not apply to them.

I took the blank CYP2D6 rows with a clearly weak screen response (weaker than -0.72, the 95th percentile of the labelled rows) and added them to the training set at $\text{pIC}_{50} = 3.5$, at most one blank row per labelled row. The screen measurement only picks the rows. It is never fed to the model as a feature, since the blind test file does not include primary screen values. The estimator is a TabPFN regressor over the organizers' `surrogate_adme` embedding.

My locked holdout could not judge this change: it only holds compounds that were hits, so it scored the pseudo-labelled arm worse on CYP2D6 (ST-RAE 0.7966 to 0.8156). I built two offline checks instead. On held-out folds, the probability that a labelled cell is predicted above a blank cell rose from 0.752 to 0.829 on development and from 0.741 to 0.824 on the holdout, while Spearman rank correlation on the labelled hits stayed level (0.4460 and 0.4499 on dev; 0.5563 and 0.5444 on holdout). Across the 750 blind compounds, the fraction of CYP2D6 predictions below 3.8 jumped from 1% to 36%.

The live board confirmed it. Uploading a file with only the CYP2D6 column swapped (`v26`) improved all five published metrics at once: macro ST-RAE went from 0.4938 to 0.4806, Spearman from 0.7234 to 0.7382, R^2 from 0.5660 to 0.5845, and rank from 55 to 43. Since the other three columns were byte-identical and the macro score is the mean of the four isoforms, the CYP2D6 column alone improved by 0.053 in ST-RAE and 0.059 in Spearman. A second upload (`v27`) raised

that arm's weight in the CYP2D6 blend from 0.4 to 0.7 (averaged over four floor and row-count settings) and reached 0.4743 at rank 41. Running the same test on CYP1A2 changed almost nothing, since the base model already separated blanks from hits at 0.86 there.

2.2 Compressing features hurts an in-context tabular model

Every TabPFN arm in this project originally projected its 605 input columns down to 200 principal components before fitting. That was a habit carried over from running on CPU, where the full width would not fit. Dropping PCA and feeding all 605 raw columns directly on GPU gave **+0.0125 of macro development Spearman**. Nothing else I tried over six weeks came close to that single jump.

The upload carrying that change improved all five blind metrics together: ST-RAE 0.5433 to 0.5333, MAE 0.7181 to 0.7094, R^2 0.5211 to 0.5317, Spearman 0.6975 to 0.7041, and Kendall 0.5224 to 0.5269. Every earlier gain had traded one metric against another, which is what happens when you shift predictions without making them better.

The memory-saving flags that forced the PCA step in the first place only cost 0.0023 on their own. The damage came from projecting 605 descriptors onto 200 components chosen to explain variance in the chemical features, which is not the same as explaining variance in CYP inhibition.

2.3 Why cluster-disjoint validation points the wrong way here

Standard advice in molecular ML is to split folds so chemical clusters never cross the split boundary, preventing the model from scoring well by memorizing close series. A competitor ranked well ahead of this entry cited cluster-disjoint splitting as the exact change that made their local numbers track the board.

Here, the measurement says otherwise. Because the organizers built the 750-compound blind set by purchasing the top 10 Tanimoto neighbours from Enamine for 75 potent hits, the blind compounds sit at a median Morgan Tanimoto similarity of **0.578** to the training set. My scaffold-disjoint holdout sits at **0.435**. Scigantic independently measured 0.587 for the blind set against 0.425 for a scaffold fold and 0.439 for a random fold.

Comparison to Training Set	Median Tanimoto	> 0.4	> 0.6
750 Blind Test Compounds	**0.578**	97.0%	40.0%
726 Scaffold-Disjoint Holdout Compounds	**0.435**	65.0%	5.5%

Table 1: Nearest-neighbour Tanimoto similarity against the 4,905 training compounds.

The scaffold holdout is already much harder than the test set. That is why holdout ST-RAE reads near 0.60 while the board returns 0.51. Filtering folds further toward strict cluster separation moves local validation away from the local-analogue regime the board actually scores.

2.4 Ensemble diversity comes from the pretraining checkpoint

I trained four message-passing networks starting from the same public Chemprop checkpoint. Between them, I varied the number of auxiliary heads (13 vs. 20), random seeds averaged (1 vs. 3), whether a ChEMBL extract was included, and whether the organizers' TDI-arm pIC_{50} columns were attached. All four models correlated **above 0.95** with each other on held-out folds. Once

any one of them was in the blend, the other three received zero weight. One of the four was the strongest single network on its own and made the combined blend worse.

A fifth network, trained with the exact same script and table but initialized from a different public checkpoint, correlated **0.70 to 0.90** with the first four. Standalone, it trailed the best network by 0.046. In the blend, it earned a 0.25 weight on three of four isoforms. Varying seeds, head counts, and auxiliary tasks produced near-duplicates; getting a genuinely different prediction required starting from a different pretraining corpus.

2.5 Recovering blind population stats from leaderboard metrics

The leaderboard prints five numbers for every upload: ST-RAE, MAE, R^2, Spearman, and Kendall. Following an idea originally published by team briford (SuperCowPowers), those observable metrics constrain three unknowns for each isoform: the blind set's mean, its spread, and the submission's correlation with it. I wrote a single-submission solver for that system and extended it to the TDI classification track. Solving those equations per isoform gave target moments to place predictions ($y \mapsto a\hat{y} + b$) without ever seeing a blind label.

3 What Did Not Work

Many ideas were run to a measured conclusion and dropped. Table 2 records the main ones.

Direction	What was run	Measured result
Public potency panels as features	Trained models on PubChem AID 1851 and Veith qHTS panels to feed their predictions into the stack.	Ranked my labels at Spearman 0.28, 0.34, 0.13, and 0.48 vs. my own models at 0.58, 0.71, 0.43, and 0.81. On CYP2D6, a model of a different isoform was the best public predictor (generic promiscuity).
Scaling the pretraining corpus	Pretrained an encoder on ∼150,000 molecules across 667 columns (ToxCast, MUV, HIV, Tox21, PubChem, ChEMBL).	Worse than an encoder trained on 33,000 molecules across 20 columns. Binary toxicity assays diluted the potency signal.
Tox21 CYP1A2 external model	Cleaned 864 luciferase counter-screen hits (AID 1224835) from AID 1671199; trained GBDT on 4,995 rows.	Reached Spearman 0.411 on challenge CYP1A2 labels vs. 0.564 for the incumbent arm; moved holdout Spearman from 0.5884 to 0.5905.
Training on the interval metric directly	Refitted models by majorize-minimize against the clipped credible interval before affine calibration.	Helped before calibration (0.7442 to 0.7244), then reversed after affine calibration (0.7198 vs. 0.7060 for standard loss). Both push predictions toward the interval edges and do not stack.
Delta learning on analogue pairs	Predicted potency differences from the closest training neighbour.	Reached 0.4488 and 0.5426 vs. 0.5881 and 0.7060 for direct arms. Blind rows sit at 0.578 similarity to train, while train rows sit far apart (pair floor 0.25).
5 multitask network variants	Added max-response heads, TDI-arm pIC_{50} heads (correlate 0.91–0.99 with direct pIC_{50}), and swapped warm-start encoders against `mtl13` (0.6262 dev).	None cleared the preset +0.015 dev gate. TDI heads scored 0.6245 dev and 0.6490 holdout (vs. 0.6652 control).
10 extra molecular encoders	Screened MiniMol, Uni-Mol, ChemBERTa, MoLFormer, and 6 others into TabPFN.	Existing encoder ranked first (0.6370 dev vs. 0.6184 for MiniMol and 0.5911 for Uni-Mol). Chemical LLMs added +0.0014 and +0.0011.
Descriptor baselines & Mitra	Tuned LightGBM, RF, SVM, and Ridge on descriptors; tested Mitra tabular foundation model.	Tree/linear models trailed the blend by 0.05–0.08. Mitra ran out of T4 VRAM and fell back to 256 examples.

Table 2: Negative results and closed directions.

4 How the Submission Was Built

4.1 The five in-house arms and the combiner

My standalone regression file (`v46`, scoring **0.4472**) combines five model families over 4,905 training compounds using five Bemis-Murcko scaffold folds.

Two arms are TabPFN regressors over 605 uncompressed columns (standardized RDKit 2D descriptors, ECFP4 counts, and frozen neural representations: one using the organizers' `surrogate_adme` encoder with the CYP2D6 blank-cell imputation, and one using a pretrained Chemprop embedding). A third arm stacks out-of-fold LightGBM predictions from the single-concentration screen.

The fourth arm (**NB33**) fine-tunes pretrained CheMeleon message-passing weights ($d_h = 2048$, depth 4) with an interval hinge on the four scored heads and MAE on 18 auxiliary heads (single-concentration $\log_2$ fold change, ChEMBL pIC_{50}, and PubChem qHTS AC_{50} and max response), following Scigantic's public 22-column table recipe on my own scaffold folds and purge rules. Under a frozen gate requiring 4/5 folds to improve and at least +0.005 dev gain, a single NB33 run gained 0.0022 and failed, while averaging three seed blocks (1042, 2042, 3042) gained 0.0062 and passed. That blend lowered CYP1A2 on the board from 0.5184 to 0.4932 in `v38`.

The fifth arm (**NB38**) was chosen by testing what the stack was missing. Instead of guessing at another architecture, I added each of Scigantic's 13 published member columns to my combiner one at a time on held-out folds. Their pretrained arms added nothing, while their randomly initialized, inverse-variance-weighted member (arm B) gained 0.019 on CYP2C9 and 0.017 on CYP2D6. I built NB38 to that spec: the NB33 network with CheMeleon initialization removed, trained on MAE across all 22 heads, with each compound weighted by the mean inverse variance of its credible intervals (normalized to the median and clipped to $[0.25, 4.0]$; 37% of the 4,179 labelled rows hit a clip). Across five folds (about five GPU hours), NB38 took the largest combiner weight on CYP2C9 and CYP2D6 and the second largest on CYP1A2, supplying roughly a third of my own-stack board improvement.

To combine all saved model columns (around 50 columns across seeds and families), I fitted an L_1-penalized linear program per isoform that minimizes the ST-RAE hinge numerator on out-of-fold rows as a correction to an incumbent column. That combiner moved the standalone stack from 0.4549 to 0.4472 (improving CYP1A2 alone by 0.0263).

For the TDI classification track (`src/tdi.py`, scored by Matthews Correlation Coefficient on CYP3A4 and CYP2D6), the pipeline combines multi-task neural shift predictions and single-concentration screen features with LightGBM classifiers, tuning probability thresholds against out-of-fold MCC for the 2-fold IC_{50} shift criterion (including inferred positives where the TDI-arm pIC_{50} exceeds 4.301 while direct $pIC_{50} < 4$).

4.2 Blending with Scigantic R25

The uploaded regression file is a weighted average of my stack (`v46`) with Scigantic's published submission R25 (github.com/Scigantic/open-admet-challenge-cyp, MIT-0 license). The weight on R25 is 0.500 for CYP1A2, CYP2D6, and CYP3A4, and 0.400 for CYP2C9, meaning **47.5% of the submitted numbers are theirs**.

Isoform	`v46` (my stack alone)	R25 (Scigantic)	`v47` (average)
CYP1A2	0.4669	0.4739	**0.4545**
CYP2C9	**0.3609**	0.3802	0.3585
CYP2D6	0.5669*	**0.5442**	0.5449
CYP3A4	0.3940	0.3762	**0.3760**

Table 3: Live leaderboard ST-RAE on 29 October before interval spread widening. (*An earlier CYP2D6 column in `v42` scored 0.5316.)

Both halves were uploaded on their own on 29 October (`v46` scored 0.4472 thirty-eight minutes after the blend `v47` scored 0.4335), so R25's contribution is measured directly at **0.0137 of macro error**. The average beats both inputs on CYP1A2, CYP2C9, and CYP3A4. On the leaderboard's secondary metrics, my standalone stack ties the blend on compound ordering (Spearman 0.7659 vs. 0.7635; Kendall 0.5868 vs. 0.5865) and loses on magnitude (MAE 0.6752 vs. 0.6547; R^2 0.5842 vs. 0.6033). My models sort the blind compounds just as well, while R25 places predictions closer to the center of the credible intervals.

My CYP2C9 column outscored R25 standalone (0.3609 vs. 0.3802). In Scigantic's build report, they noted a mismatch in their R24 step: out of fold, their Powell combiner fitted member differences against an R08-like baseline, then applied those weights on the blind set to R17's CYP2C9 column (which came from R10). Giving my stack 0.600 weight on CYP2C9 leaned on the stronger column.

4.3 Interval placement and spread calibration

Scigantic's combiner ends every step with a rank-preserving `restore()` call that resets the column's mean and standard deviation back to their R17 baseline. That prevented blending from shrinking their spread, and it also left overall location and width untuned.

In the training data, weak CYP2D6 compounds ($\text{pIC}_{50} < 3.6$) have a median upper credible bound of 3.75, whereas my raw CYP2D6 predictions had a median of 3.20 with 65% of values below 3.5. Because development folds have no labels for true non-inhibitors, the only way to measure weak-compound placement was on the board one column at a time. Pulling CYP2D6 values below 3.5 upward via $x \mapsto 3.5 - g(3.5 - x)$ scored **0.5524** at $g = 0.5$, 0.5556 at $g = 0.3$, and 0.5714 at $g = 0.15$. Applying a similar floor at 3.6 across all four columns at once (`v39`) hurt every tab.

Between 29 October (0.4335) and the final upload (`v56`, **0.4192**), the remaining gain came from widening each isoform's spread around its mean along an affine transform that preserves both the column mean and every compound's rank order, with no change to any underlying model. About 39% of that move (and roughly half of the total calibration gain across the project) is reproduced by a single flat 1.2× widening applied to all four columns, which compensates for ensemble predictions being narrower than the assay spread.

5 Where Validation Misled Me

A locked holdout of 726 scaffold-disjoint compounds was scored once per candidate. Across three early uploads, the offset between holdout ST-RAE and leaderboard ST-RAE was −0.0834, −0.0728, and −0.0833. Two of the three agreed to within 0.0001, and using that offset predicted the third upload onward to within 0.0004.

Two exceptions mattered. First, one candidate improved the holdout by 0.0046 (with a paired scaffold bootstrap giving a 0.90 probability of improvement) and then scored 0.0060 worse on the live board. A 726-compound holdout cannot resolve effects of 0.005, and several early gates were written as though it could.

Second, CYP2D6 failed to transfer from local validation to the board four separate times: the pseudo-label arm hurt the holdout by 0.019 and gained 0.053 on the board; upweighting it hurt the holdout by 0.021 and gained 0.025 on the board; NB33 gained 0.0172 on the holdout and lost 0.007 on the board; and the combiner predicted 0.5282 for the blend where the board returned 0.5449. Because the blind set was expanded around 1A2, 2C9, and 3A4 hits rather than 2D6 hits, the blind CYP2D6 column is dominated by weak compounds, while the labelled holdout contains only active hits. For any change aimed at the inactive distribution, a hit-only holdout tests the wrong population.

6 Limitations and Attribution

Without Scigantic's R25 predictions, my standalone stack (`v46` at 0.4472) sits outside the top ten percent of the live leaderboard. Scigantic's own live standing on 28 October was 0.4283 (from a later upload where they added flat constants of +0.170 to CYP1A2 and +0.140 to CYP2C9 on top of R25), so `v56` (0.4192) sits ahead of their best reading by 0.0091, a margin produced by post-blend spread calibration rather than by the models themselves. Entries ranked near the top of the board run six base model families against my three and pretrain backbones across much larger bioactivity tables. Finally, every finding here is measured on one dataset of 4,905 training compounds and 750 blind compounds, where the live board scores half of the test set (split by parent chemical series) and has a single-isoform noise floor around 0.002.

Component	Source	Measured effect
5-arm stack over 4,905 training compounds	This entry	0.4472 alone
Blank cells read as non-inhibitors + CYP2D6 pseudo-label arm	This entry	Rank 55 to 41 (`v26`/`v27`)
CYP2D6 floor compression (standalone pre-`v56` test)	This entry	0.5524 on CYP2D6
The CYP2C9 column	This entry	Outscored R25 alone
L_1 linear program combiner over ∼50 saved model columns	This entry	0.4549 to 0.4472
Reading the board one column at a time + frozen dev gate	This entry	4 measurements/upload
Single-submission metric inversion & TDI track extension	This entry	Distribution bounds
R25 predictions (0.500 on 1A2/2D6/3A4, 0.400 on 2C9)	Scigantic (MIT-0)	−0.0137 macro error
22-column training table & CheMeleon/Arm-B recipes	Scigantic (MIT-0)	Used in NB33/NB38
Affine placement correction & leaderboard probing idea	briford (SuperCowPowers)	Centering step
Chemprop, TabPFN, and CheMeleon checkpoints	Original authors	Base backbones

Table 4: Who made which part.

Code and reproduction scripts are in this repository. The full experiment log is in `STATUS.md` and

METHOD.md.